UNIVERSITY PROJECT PROPOSAL

Store Sales Dataset Analysis Using Excel & Power BI

Microsoft Excel · Power BI · Power Query · DAX | 7 Weeks

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Project Overview

PROPOSAL

1

Overview

Analyze store sales data with Excel and Power BI to find sales trends, evaluate performance, understand customers and generate actionable insights.

2

Problem Statement

Raw sales data alone gives no clear view of performance. It must be cleaned, analyzed and visualized to support better business decisions.

Project Objectives

PLANNED

1. Clean and prepare the dataset with Excel and Power Query
2. Analyze sales, profit, products, customers, regions and shipping
3. Develop and validate DAX measures
4. Build interactive Power BI dashboards
5. Identify trends, patterns and business opportunities
6. Provide data-driven recommendations

Dataset & Methodology

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Superstore sales dataset (project repository). Suitability to be confirmed in Week 1; the 44 measures are subject to it.

1. **Data Understanding**
2. **Data Cleaning**
3. **Exploratory Analysis**
4. **Modeling & DAX**
5. **Dashboard Development**
6. **Validation & Insights**
7. **Final Delivery**

Excel Cleaning & Exploratory Analysis

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Data Cleaning (Week 2)

- Check missing values
- Remove duplicates
- Validate data types
- Fix inconsistencies
- Document procedures

Exploratory Analysis (Week 3)

- Pivot Tables
- Charts
- Sales, profit, discounts
- Customers and products
- Record initial findings

44 Planned DAX Measures (1–24)

PLANNED

Core Sales & Revenue KPIs (8)

1. Total Sales
2. CY Sales (Current Year Sales)
3. LY Sales (Last Year Sales)
4. Average Daily Sales
5. Average Monthly Sales
6. Maximum Sale Value
7. Minimum Sale Value
8. Total Transactions Count

Customer & Segment Analytics (8)

9. Total Unique Customers
10. New Customers Count
11. Repeat Customers Count
12. Average Sales Per Customer (ARPU)
13. Corporate Segment Sales
14. Consumer Segment Sales
15. Home Office Segment Sales
16. Consumer Sales Share %

Time Intelligence (8)

17. Sales YTD (Year To Date)
18. Sales MTD (Month To Date)
19. Sales QTD (Quarter To Date)
20. YoY Sales Growth %
21. MoM Sales Growth %
22. Sales PM (Previous Month)
23. Sales PQ (Previous Quarter)
24. YoY Sales Variance

44 Planned DAX Measures (25–44)

PLANNED

Product & Category Performance (8)

25. Total Products Sold Quantity
26. Unique Products Count
27. Furniture Category Sales
28. Office Supplies Category Sales
29. Technology Category Sales
30. Technology Sales Share %
31. Product Sales Rank
32. Top 10 Products Flag

Geographic & Shipping Analytics (8)

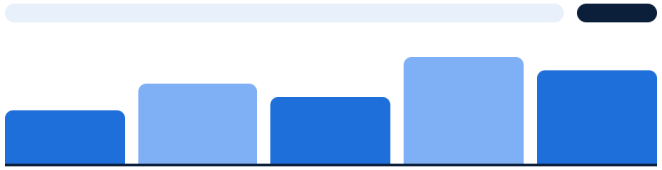
33. Total Orders
34. Average Shipping Days
35. First Class Shipments Count
36. Standard Class Shipments Count
37. Express Shipping Share %
38. Regional Sales
39. Active Cities Count
40. Average Sales Per City

Advanced & Dynamic Display (4)

41. Dynamic Title Measure
42. Conditional Formatting Growth Color
43. Top Customers Filter Measure
44. Selected Period Sales

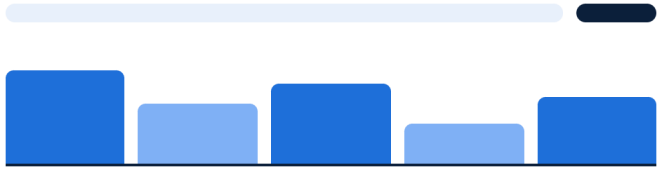
Six Proposed Dashboard Pages

PLANNED



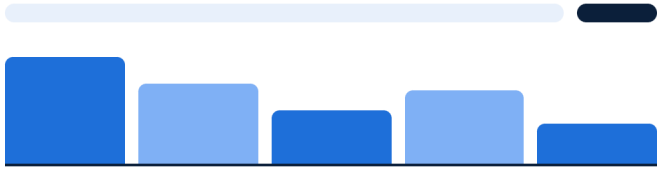
Sales Overview

KPIs and sales trends



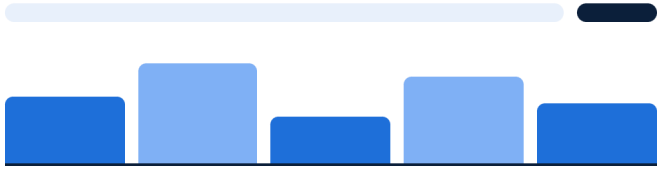
Customer Analysis

Customer segments



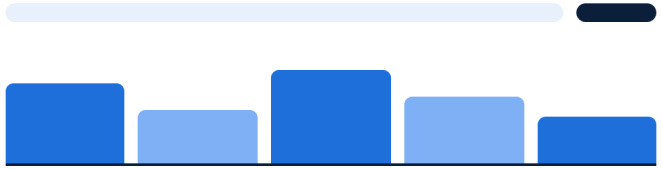
Product Analysis

Products and profit



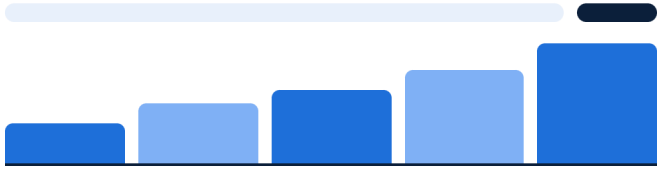
Geographic Analysis

Regional performance



Shipping Analysis

Shipping performance



Time Analysis

Trends over time

Illustrative layouts only. No results are shown.

Seven-Week Timeline

PLANNED

1

Week 1

Setup & Data Understanding

Deliverable: Scope and dataset overview

2

Week 2

Cleaning & Preparation

Deliverable: Clean dataset and cleaning report

3

Week 3

Exploratory Analysis

Deliverable: Excel analysis, initial insights

4

Week 4

Modeling & DAX

Deliverable: Data model, validated DAX

5

Week 5

Dashboard Development

Deliverable: Initial interactive dashboard

6

Week 6

Testing & Insights

Deliverable: Validated dashboard, insights report

7

Week 7

Finalization & Presentation

Deliverable: Final dashboard, report, slides

Team Roles & Responsibilities

Mahmoud Walid Elsayed Mohammed

Team Leader & Project Coordinator

Objectives, repository, progress, integration and project documentation

Mostafa Khaled Zakaria Abdellatif

DAX & Analytics Developer

Plans, documents and tests the 44 DAX measures

Habiba Abdelaziz Waheid Abdelaziz

Data Analysis & Business Insights Specialist

Trends, insights and recommendations; verifies calculations and results

Salma Mohammad Akl

Dashboard Designer

Power BI pages, charts, cards, slicers and navigation; tests filters and functionality

Expected Deliverables & Outcomes

PLANNED

Deliverables

- Cleaned dataset
- Excel analysis workbook
- Power BI model and DAX measures
- Interactive dashboard
- Insights and recommendations report
- Documentation and presentation

Business Outcomes

- Clear view of store performance
- Sales and profit trends
- Key customer and product segments
- Data-driven decision support

Conclusion

Excel and Power BI will turn raw store sales data into useful business insights, through a structured seven-week workflow with defined responsibilities and deliverables.

Status: proposal stage. All activities are planned; none are completed.

Thank you · Questions